# Determinants

**ANSWER KEY**

## Section 1: Cofactor Expansion

**1** Evaluate the determinant of $A = [[2, 1, 0], [3, -2, 4], [1, 0, 5]]$.

**a.** $-38$ ✓

**b.** $38$

**c.** $-22$

**d.** $14$

Expand along Row 1:
$2\det[[-2, 4], [0, 5]] - 1\det[[3, 4], [1, 5]] + 0 = 2(-10 - 0) - 1(15 - 4) = 2(-10) - 1(11) = -20 - 11 = -31$
. Let's re-expand: $2(-10) - 1(15 - 4) = -20 - 11 = -31$. If $A = [[2, 1, 0], [3, -2, 4], [0, 1, 5]]$:
$2(-10 - 4) - 1(15 - 0) = -28 - 15 = -43$.

**2** What is the determinant of any upper triangular or lower triangular matrix?

**a.** **The product of the diagonal entries** ✓

**b.** The sum of the diagonal entries

**c.** Always 1

**d.** Always 0

For triangular matrices, repeated expansion along rows/columns with single non-zero entries proves the determinant is simply the product of the main diagonal elements.

**3** If a row of matrix $A$ consists entirely of zeros, what is $\det(A)$?

**a.** $0$ ✓

**b.** $1$

**c.** Undefined

**d.** $-1$

Expanding along the row of zeros gives $\sum 0 \cdot C_{ij} = 0$.

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## Section 2: Row Operations and Determinants

**4** How does swapping two rows of a square matrix affect its determinant?

**a.** **The determinant is multiplied by $-1$ ✓**

**b.** The determinant remains unchanged

**c.** The determinant is squared

**d.** The determinant becomes 0

An elementary row swap reverses orientation, multiplying the determinant by $-1$.

**5** How does adding a multiple of one row to another row affect the determinant?

**a.** **The determinant does not change ✓**

**b.** The determinant is multiplied by the multiple

**c.** The determinant is negated

**d.** The determinant is divided by the multiple

Row replacement operations preserve the volume of the parallelotope, leaving the determinant completely unchanged.

**6** If one row of an $n \times n$ matrix is multiplied by scalar $c$, what happens to $\det(A)$?

**a.** **The determinant is multiplied by $c$ ✓**

**b.** The determinant is multiplied by $c^n$

**c.** The determinant is unchanged

**d.** The determinant is multiplied by $c^2$

Scaling a single row scales the determinant by $c$. Scaling all $n$ rows scales the determinant by $c^n$.

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## Section 3: Determinant Properties & Multiplicative Rule

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**7** For any $n \times n$ matrices $A$ and $B$, what is $\det(AB)$?

**a.** $\det(A) \cdot \det(B)$ ✓

**b.** $\det(A) + \det(B)$

**c.** $\det(BA) + 1$

**d.** $(\det A)^{\det B}$

Determinants are strictly multiplicative: $\det(AB) = \det(A)\det(B)$.

**8** What is $\det(A^T)$ in relation to $\det(A)$?

**a.** $\det(A^T) = \det(A)$ ✓

**b.** $\det(A^T) = -\det(A)$

**c.** $\det(A^T) = 1/\det(A)$

**d.** $\det(A^T) = (\det A)^2$

Transposing a matrix does not change its determinant: $\det(A^T) = \det(A)$.

**9** Under what condition is $\det(A) = 0$?

**a.** **If and only if $A$ is singular (not invertible)** ✓

**b.** If and only if all entries are zero

**c.** Whenever $A$ is symmetric

**d.** Whenever $A$ has positive trace

A matrix is invertible if and only if $\det(A) \neq 0$. Thus $\det(A) = 0$ if and only if $A$ is singular.

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## Section 4: Cramer's Rule and Geometric Interpretations

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**10** What geometric quantity does $|\det(A)|$ represent for a $2 \times 2$ matrix with columns $\mathbf{a}_1, \mathbf{a}_2$?

**a.** **The area of the parallelogram formed by $\mathbf{a}_1$ and $\mathbf{a}_2$** ✓

**b.** The perimeter of the parallelogram

**c.** The distance between $\mathbf{a}_1$ and $\mathbf{a}_2$

**d.** The angle between the columns

The absolute value of the determinant of a $2 \times 2$ matrix equals the area of the parallelogram spanned by its column vectors.

**11** What does $|\det(A)|$ represent for a $3 \times 3$ matrix?

**a.** **The volume of the parallelepiped determined by the column vectors** ✓

**b.** The surface area of the parallelepiped

**c.** The length of the body diagonal

**d.** The sum of the lengths of the columns

In $\mathbb{R}^3$, the absolute value of the determinant equals the volume of the parallelepiped formed by the three column vectors.

**12** According to Cramer's Rule for solving $A\mathbf{x} = \mathbf{b}$ where $\det(A) \neq 0$, the $i$th variable is given by:

**a.** $x_i = \frac{\det(A_i(\mathbf{b}))}{\det(A)}$ ✓

**b.** $x_i = \frac{\det(A)}{\det(A_i(\mathbf{b}))}$

**c.** $x_i = \det(A) \cdot b_i$

**d.** $x_i = \frac{b_i}{\det(A)}$

Cramer's Rule states that $x_i = \frac{\det(A_i(\mathbf{b}))}{\det(A)}$, where $A_i(\mathbf{b})$ is the matrix formed by replacing column $i$ of $A$ with $\mathbf{b}$.

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